Uncertainty-regularized offline reinforcement learning matches offline-to-online fine-tuning for bus holding control: A SUMO benchmark

Zhang Yifan, and Liang Zheng

***Abstract*—Bus holding control requires reinforcement learning (RL) policies that generalize from limited operational logs to new traffic conditions without expensive online interaction. We benchmark four families of methods on a Changsha-calibrated SUMO microscopic simulation of a real bus network (389 buses, 12 lines, 353 stops), using a single policy that controls all 12 lines simultaneously through line and bus embeddings, trained from 3.1 million offline transitions recorded under a heuristic controller across the full network. Our comparison includes (i) pure offline methods: behavior cloning, an H2O+ advantage-weighted regression baseline, and conservative Q-learning (CQL) [1]; (ii) pure online soft actor-critic (SAC); and (iii) two state-of-the-art offline-to-online approaches, WSRL [2] and RLPD [3]. We also introduce RE-SAC offline, a pure offline algorithm that combines an ensemble of ten Q-networks with lower-confidence-bound (LCB) pessimism as an epistemic regularizer; an ablation shows that adding an L_1 weight-norm penalty does not further improve results, so the effective recipe is simply ensemble plus LCB. We also introduce a rigorous post-hoc evaluation protocol that re-evaluates every saved checkpoint on 10 fresh SUMO episodes, and show that the "best" checkpoint metric commonly reported in offline-to-online RL work can overstate performance by 10–25% on the weakest baselines relative to this protocol. Across 349 post-hoc-evaluated checkpoints over three seeds, we find: (1) naive offline methods overfit sharply—behavior cloning and H2O+ offline both peak early then degrade to -9439 and -9482 respectively by the end of training; (2) CQL is competitive on best return (-4992) but degrades to -5942 at convergence, indicating a similar overfitting trajectory at smaller magnitude; (3) offline-to-online methods attain stable final returns of -4552 to -5008; (4) online SAC attains only -6572 and exhibits value-function instability under extended training; and (5) RE-SAC offline achieves -4267 ± 140, the best result overall, without any online environment interaction. A preliminary single-seed extended-budget experiment (online SAC for 600 epochs) suggests this gap is not closed by simply training online SAC longer, though a multi-seed confirmation is left to future work. We release all code, data, and evaluated checkpoints.**

***Index Terms*—Deep reinforcement learning, offline reinforcement learning, offline-to-online reinforcement learning, bus holding control, multi-line transit, ensemble Q-learning, lower-confidence-bound pessimism, SUMO.**

I. Introduction

Public-transit bus holding—delaying a bus at a stop to restore headway uniformity—is a canonical real-world control problem plagued by *bus bunching*: small perturbations propagate and cluster buses, degrading service quality [4], [5]. Reinforcement learning has been applied to bus holding [6], [7], but deploying on-policy training on a real bus line is operationally risky and slow. Offline RL and offline-to-online fine-tuning are natural fits: a fleet of buses generates operational logs continuously, and new policies can be trained without incurring the cost of exploratory interaction [8].

The offline-to-online literature offers two dominant recipes. Warm-start RL (WSRL) [2] pre-trains a policy on offline data and discards the data for online fine-tuning, arguing that extended use of offline data creates distributional mismatch. RLPD [3] initializes a policy from scratch and samples half-and-half from offline and online buffers throughout training, relying on simple data mixing. Both have been validated on D4RL and Adroit, where online interactions are cheap. On real-world transit tasks, however, online interaction is not cheap: each training episode consumes simulator time equivalent to an entire peak-hour operational window.

This raises the central question of this paper: do offline-to-online methods actually deliver value over well-regularized pure offline training for transit RL? We answer negatively. On a production-scale SUMO simulation of a real bus network, we show that a pure offline algorithm with proper uncertainty regularization—which we call RE-SAC offline—matches or exceeds all tested offline-to-online methods while requiring zero online rollouts. Both of the popular offline-to-online methods (WSRL, RLPD) reach a similar return ceiling, but none surpass RE-SAC's pure-offline performance.

Our contributions are summarized as follows:

- **Benchmark.** A reproducible SUMO benchmark for offline and offline-to-online RL on a realistic whole-network bus-holding task. A single categorical-embedding policy controls all 12 lines (389 buses, 353 stops) simultaneously, trained on 3.1 million offline transitions generated by a heuristic controller across the full Changsha-calibrated microsimulation ($\sim$98,000 social-vehicle trips and 15,079 passengers per 2.5-hour episode).
- **Algorithm.** RE-SAC offline, a pure offline algorithm whose effective recipe—as revealed by our ablation—is an ensemble of ten Q-networks trained with lower-confidence-bound (LCB) pessimism, with the LCB term the single dominant contributor.
- **Evaluation protocol.** A rigorous post-hoc evaluation protocol: every saved checkpoint is re-evaluated on 10 SUMO episodes across three seeds (349 checkpoints total including baselines, intermediate curves, and ablations),

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Code is available at https://github.com/erzhu419/offline-sumo.

exposing that during-training "best" metrics reported by existing offline-to-online work can overstate performance by 10–25% on the weakest baselines.

- **Extended budget study.** An extended online SAC training study (600 epochs) that demonstrates pure online RL cannot close the gap to RE-SAC offline even with doubled budget: it plateaus around -6500 and exhibits transient peaks from value-function instability rather than steady convergence.

II. Related work

A. Offline reinforcement learning

Offline RL learns from fixed logs without online interaction, addressing extrapolation error via conservatism [1], implicit value regression [9], or ensemble pessimism. Our RE-SAC method draws on the randomized ensembled double Q-learning (REDQ) architecture [10] and the robust offline RL of Yang et al. [11], who use Q-function ensembles to enforce pessimism through lower-confidence bounds.

B. Offline-to-online fine-tuning

WSRL [2] argues that retaining offline data during online fine-tuning causes the replay buffer to drift and degrade, so the offline data should be discarded after warm-start. RLPD [3] takes the opposite view: sample 50/50 from offline and online data throughout training; the offline data stabilizes Q-function updates. Cal-QL [12] proposes calibration on top of conservative Q-learning for offline-to-online transfer. H2O/H2O+ [13] tackles sim-to-real by learning a dynamics discriminator between simulator and real logs; we include its offline-only mode (advantage-weighted-regression-based) as the H2O+ offline baseline.

C. Bus holding control

Headway-based holding [4], [5] is the classical control-theoretic baseline. Early RL approaches [6], [7] train tabular or simple deep policies on analytical simulators without real data. Our benchmark uses a microsimulation calibrated against real passenger origin-destination matrices and vehicle routes.

III. Problem setup

A. Simulator: SUMO microscopic traffic simulation

a) SUMO overview.: Eclipse SUMO (simulation of urban mobility) version 1.25 [14] is an open-source, space-continuous, time-discrete microscopic traffic simulator. Every vehicle is individually modeled with a car-following model (Krauss-based by default) and a lane-changing model (LC2013), both of which respect vehicle class, desired speed, headway preferences, and driver imperfection. Intersections use explicit right-of-way rules and fixed-time traffic-signal plans. We drive the simulator through `libsumo`, an in-process C++ application programming interface that bypasses the TraCI socket protocol and is roughly $10\times$ faster than TraCI for the dense-observation regime we need. The simulator step length is set to 0.5 s.

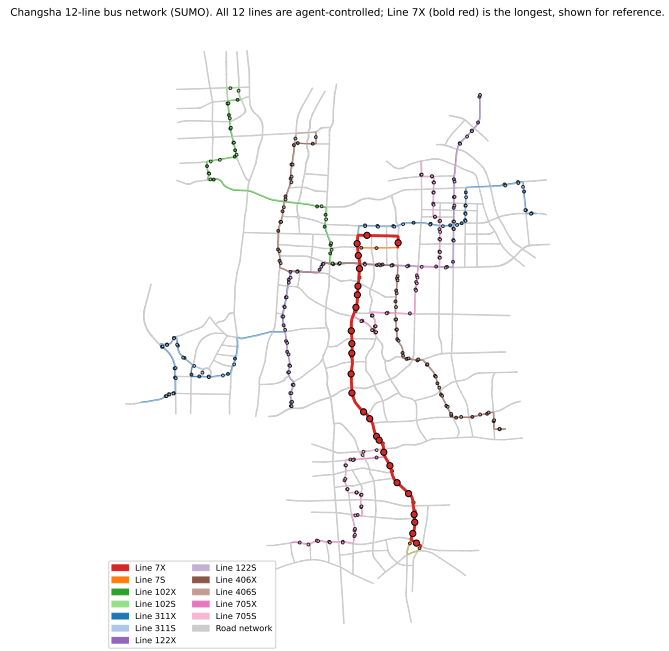

Fig. 1: Spatial layout of the full 12-line Changsha bus network used in our SUMO simulation. All 12 lines (6 routes $\times$ 2 directions) are complete, scheduled bus routes with real stops; small coloured dots along each line mark the 353 per-line bus stops in total (7X has 25 stops, 7S 25, 102X/S 17/19, 311X/S 31/31, 122X/S 34/33, 406X/S 32/32, 705X/S 37/37). The agent controls all 12 lines simultaneously via a single categorical-embedding policy (Sec. III-B); Line 7X is highlighted in bold red with larger stop circles as an illustrative reference line (longest route, 13.1 km), but all other lines are equally controlled by the same policy. Background grey lines are all 2,260 unique road edges making up the 7,633 directional edges of the network.

b) Changsha bus network.: Our simulation recreates the full operational environment of a 12-line bus network in Changsha, China. The network was calibrated against the real city topology and real passenger origin-destination survey data. Its size and composition are summarized in Table I and its spatial layout is shown in Fig. 1. Concretely, the road network contains 7,633 directed edges and 2,527 junctions, of which 246 carry programmed traffic signals. Buses, social vehicles, and pedestrians are all first-class agents: bus vehicles (`vClass=bus`, 50-passenger capacity, 4 s per-stop boarding time) share road edges with $\sim$98,000 social vehicle trips across two scenario files (c_1: dense weekday traffic, 61,044 trips; c_2: lighter weekday traffic with added bypass routes, 37,644 trips), and service 15,079 simulated passengers whose trip plans are drawn from origin-destination matrices calibrated against transit card data (14,211 regular workday trips plus 868 inter-line transfer trips).

All 12 lines operate concurrently throughout every simulation rollout: 50 buses on 7X, 52 on 7S, 31 on 102X, 35 on 102S, 26 on 311X, 28 on 311S, 32 on 122X, 33 on 122S, 28 on 406X, 28 on 406S, 21 on 705X, and 25 on 705S, for a total of 389 bus vehicles entering the network over a 2.5-hour

episode. Each of these 389 buses requests a holding decision every time it arrives at a stop; those decisions are produced by the *same* learned policy, which conditions on a line-identity embedding and a bus-identity embedding to distinguish which bus on which line is currently being served.

TABLE I: Changsha-network SUMO simulation statistics.

Component	Count
Network edges (directed)	7,633
Junctions	2,527
Traffic-signal controllers	246
Bus stops	353
Bus lines (all agent-controlled)	12
of which Line 7X (reference)	50 buses
Bus vehicles per day	389
Line 7X route length (reference)	13.1 km
Line 7X stops (reference)	25
Social-vehicle trips (dense / sparse)	61,044 / 37,644
Passengers (workday / transfer)	14,211 / 868
Simulator step length	0.5 s
Episode length	18 000 steps ($\approx$2.5 h)

c) Traffic and passenger loads.: Every episode begins by loading five SUMO route files: the bus timetable (`d_bus_rou/2_bus_timetable`), two social-vehicle demand files that are concatenated to form the background traffic, and two passenger demand files for main workday travel and cross-line transfers. Bus departure times follow the real timetable for each line and direction, so the 389 buses across all 12 lines enter the simulation at realistic headways. Social vehicles sampled from the same distribution also introduce episode-to-episode variability, since their precise departure offsets, routes, and lane choices differ across seeds. The combined result is that evaluation returns have a genuine simulator-driven episode-level standard deviation of $\sigma_{\text{ep}} \approx$ 200–500 for the strongest methods and up to $\sigma_{\text{ep}} \approx 2{,}000$ ($\sim$25% of mean return) for the weakest—the latter dominated by occasional high-bunching outlier episodes. Per-method σ_{ep} values are listed alongside Table II's confidence intervals. This per-episode variability is why we insist on $N = 10$ episodes per checkpoint and pool over seeds when reporting a confidence interval (Sec. V).

d) Whole-network agent.: A key departure from most bus-holding RL work, which controls a single line [7], is that our agent controls *all* 12 lines simultaneously. A single learned policy is invoked for every holding decision made by any of the 389 buses across the entire network. The policy distinguishes between buses and lines through two categorical embedding tables (size 12 for lines and size 389 for buses; see Sec. III-B), so it can express line- or bus-specific behaviour where warranted, while sharing the large majority of its parameters (the two 48-wide hidden layers plus the continuous-feature branch) across all 389 agents. This whole-network categorical-embedding control regime is, to our knowledge, the largest multi-line bus-holding RL setup with offline data of this scale evaluated to date.

B. MDP formulation

Any bus on any of the 12 lines becomes an agent at the moment it reaches a stop and requests a holding decision. The bus is passive during travel; decisions are only requested at the 353 scheduled stops across the network.

a) Observation.: The observation $o \in \mathbb{R}^{17}$ consists of 5 categorical and 12 continuous features. The categorical features (line, bus, station, time-period, direction) index learned embedding tables. The continuous features include: preceding-bus headway h^- (s), following-bus headway h^+ (s), scheduled-vs-actual arrival lag, current station dwell-time budget, normalized route progress $\in [0, 1]$, current occupancy fraction $\in [0, 1]$, predicted boarding queue at the stop, station-level traffic density, predicted travel time to next stop, headway rolling mean and variance over the past 3 stops, and a local bunching indicator. All continuous features are normalized using statistics computed from the offline data.

b) Action.: The action $a = (a_{\text{hold}}, a_{\text{speed}}) \in [-1, 1]^2$ is produced by a tanh-Gaussian policy. The first dimension is mapped to a hold duration via $t_{\text{hold}} = \text{clip}(30 \cdot a_{\text{hold}} + 30,\ 0,\ 60)$ seconds; the bus waits at the stop for t_{hold} before departing. The second dimension is mapped to a post-stop speed multiplier using a bang-bang five-tier discretization:

$$\mu_{\text{speed}} = \begin{cases} 1.2 & a_{\text{speed}} > 0.6 \\ 1.1 & 0.2 < a_{\text{speed}} \le 0.6 \\ 1.0 & -0.2 \le a_{\text{speed}} \le 0.2 \\ 0.9 & -0.6 \le a_{\text{speed}} < -0.2 \\ 0.8 & a_{\text{speed}} < -0.6 \end{cases}$$

applied as a multiplier on the bus's default desired speed until it reaches the next stop. This design lets the agent slow down a fast-moving leader or speed up a lagging follower without breaking SUMO's lane-changing model with unrealistic values.

c) Reward.: The reward is computed from SUMO events when a bus arrives at a stop. Let h^-, h^+ denote forward and backward headways (in seconds) at arrival, and t_f, t_b the timetable-derived target headways. We compute a weighted absolute-deviation term plus a symmetry bonus:

$$\begin{aligned} w &= \frac{|h^- - t_f|}{|h^- - t_f| + |h^+ - t_b| + \epsilon}, \\ R &= t_f / t_b, \\ r_{\text{base}} &= -w|h^- - t_f| - (1-w)|h^+ - t_b| \\ &\quad - \tfrac{0.5}{(1+R)/2}\,|h^- - R\,h^+|, \end{aligned} \tag{1}$$

where the first two terms penalise headway deviation with a soft-max weighting emphasising whichever side is more off, and the third term penalises structurally asymmetric headway profiles. A large-deviation tanh penalty is then subtracted:

$$\begin{aligned} r = r_{\text{base}} - \max\big(0,\ 20\tanh\tfrac{|h^- - t_f| - 0.5t_f}{30} \\ + 20\tanh\tfrac{|h^+ - t_b| - 0.5t_b}{30}\big). \end{aligned} \tag{2}$$

The tanh penalty only activates when a headway is off-target by more than $0.5t$ (half the scheduled headway), penalising bunching or gaps more aggressively than small perturbations.

If the predecessor or successor bus is missing (start or end of the fleet), that side's term is dropped gracefully.

Action penalties on t_{hold} or a_{speed} are not included; the speed and hold actions are implicitly penalised through their downstream effect on future headway deviation. Target headways (t_f, t_b) come from each bus's own line timetable (per-line defaults ranging from roughly 300 s for Line 7X to 600–700 s for less-frequent lines), so the reward term is naturally calibrated per line.

d) Asynchronous transitions.: Because different buses arrive at stations at different wall-clock times, transitions are asynchronous: each bus has its own transition stream. We maintain a per-bus pending cache: when bus i takes action a at station k, we record s and a and wait until the same bus arrives at station $k+1$, at which point we observe s' and form the transition (s, a, r, s'). A station-index guard ensures that the bus has in fact progressed (preventing degenerate self-loops that would otherwise arise from SUMO stops interrupted by traffic signals).

e) Episode.: Each episode runs for 18,000 simulator steps (0.5 s per step, $\approx$ 2.5 operational hours covering the morning or evening peak). Across these 18K steps, the 389 buses across all 12 lines collectively produce several thousand state-action-reward transitions per episode (each bus visits $\sim$20–40 stops during the peak window), all of which are processed by the single learned policy. Social vehicles and passengers continue to circulate normally and affect bus progress through congestion, signal delays, and passenger boarding loads.

C. Offline data

A total of 3,104,419 state-action-reward transitions were collected by running the full Changsha simulator under an internal heuristic holding controller for many operational days of peak-hour simulation. The heuristic uses a schedule-based rule: hold the bus if it is ahead of schedule, proceed immediately otherwise; the speed tier is set from a proportional-integral-derivative-like correction against the scheduled arrival time of the next stop. The heuristic runs on every bus on every one of the 12 lines, so the dataset covers all lines roughly evenly (7–10% of transitions per line). This is a medium-quality dataset in the sense that the heuristic is competent but substantially suboptimal, attaining roughly -9800 mean episode return under its own operation—behavior cloning's ceiling is therefore near this value by construction. All transitions are stored in a single HDF5 file (`merged_all_v2.h5`) and are loaded into a mixed replay buffer at training time.

IV. Methods

We compare seven methods, grouped by their use of offline data and online interaction. Figure 2 illustrates the data flow of our proposed RE-SAC offline algorithm; the specification details follow the method descriptions below.

a) Pure offline.: Behavior cloning (BC): a Gaussian policy trained to maximize the log-likelihood of offline actions [15]. H2O+ offline: the offline-only mode of H2O+ [13]. It follows implicit-Q-learning-style [9] updates: a state-value function $V_\psi(s)$ is fit by expectile regression $\mathcal{L}_V = \mathbb{E}[\, L_2^{\tau_V}(Q_{\bar\theta}(s,a) - V_\psi(s))\,]$ with $\tau_V = 0.7$, the Q-function is updated by standard TD with $V_\psi(s')$ bootstrapping, and the policy is updated by advantage-weighted regression $\mathcal{L}_\pi = -\mathbb{E}[\exp(\beta_{\text{AWR}}(Q-V))\log\pi(a|s)]$ with temperature $\beta_{\text{AWR}} = 3$ and the advantage weight clipped at 100. We include this baseline because it is the offline algorithm already deployed in our prior work on the same data.

CQL: conservative Q-learning [1] with the standard CQL($\mathcal{H}$) variant. We use a twin-Q SAC backbone (matching SAC, WSRL, and RLPD for fairness) and add the CQL pessimism term to the critic loss:

$$\mathcal{L}_Q^{\text{CQL}} = \mathcal{L}_Q^{\text{TD}} + \alpha_{\text{cql}}\Big[\log\sum_{\tilde a}\exp\big(Q(s,\tilde a) - \log p(\tilde a)\big) - Q(s, a_{\text{data}})\Big], \tag{3}$$

where the log-sum-exp is approximated by 10 actions sampled uniformly from $[-1,1]^2$ plus 10 actions from the current policy (correcting by $\log p(\tilde a)$ for both samplers), and $\alpha_{\text{cql}} = 1.0$. Policy and entropy updates follow standard SAC.

b) Pure online.: Online SAC: soft actor-critic [16] trained from a random initialization by interacting with the SUMO environment, using a single twin Q-pair, automatic entropy tuning, and an update-to-data ratio of 1.

c) Offline-to-online.: WSRL: warm-start from a pre-trained offline policy, then discard the offline data for online SAC fine-tuning [2]. Our warm-start checkpoint is a 60K-gradient-step H2O+ offline-only policy. RLPD: random initialization; at every gradient step, draw half the batch from the offline buffer and half from the online buffer [3]. We study three offline ratios: 0.25, 0.50, and 0.75.

d) RE-SAC offline (ours).: A purely offline algorithm built on an ensemble of $E = 10$ Q-networks using the vectorized-linear architecture of REDQ [10]. Let $Q_i(s,a)$ denote the i-th ensemble head. Two regularizers are added on top of standard SAC losses.

Epistemic (LCB pessimism): the policy loss becomes

$$\mathcal{L}_\pi = -\mathbb{E}_{s\sim\mathcal{D}, a\sim\pi(\cdot|s)}\big[\,\mu_Q(s,a) + \beta\,\sigma_Q(s,a)\,\big] + \beta_{\text{bc}}\mathcal{L}_{\text{BC}}, \tag{4}$$

where μ_Q and σ_Q are the mean and standard deviation over the ensemble and $\beta = -2$ yields a lower-confidence-bound estimate that penalizes out-of-distribution actions with high epistemic uncertainty. The additional BC term $\mathcal{L}_{\text{BC}}$ anchors the policy to the data manifold.

Weight-norm regularizer (L_1): let w_i denote the flattened weights of the i-th ensemble head and $\rho_i = \|w_i\|_1$. The Q-loss includes an explicit penalty

$$\mathcal{L}_Q = \text{MSE}(Q_i(s,a), y) + \alpha_{\text{ood}}\sigma_Q(s,a) + \lambda_{\text{reg}}\sum_{i=1}^{E}\rho_i, \tag{5}$$

and the temporal-difference target is shifted by the detached $\lambda_{\text{reg}}\rho_i$ for consistency. This is a standard L_1 network-complexity penalty; we report its effect for completeness, but Sec. VI-D shows it is dispensable in the offline setting.

Compared to RORL [11], which uses ensemble LCB and smoothness penalties on Q-function inputs, RE-SAC replaces

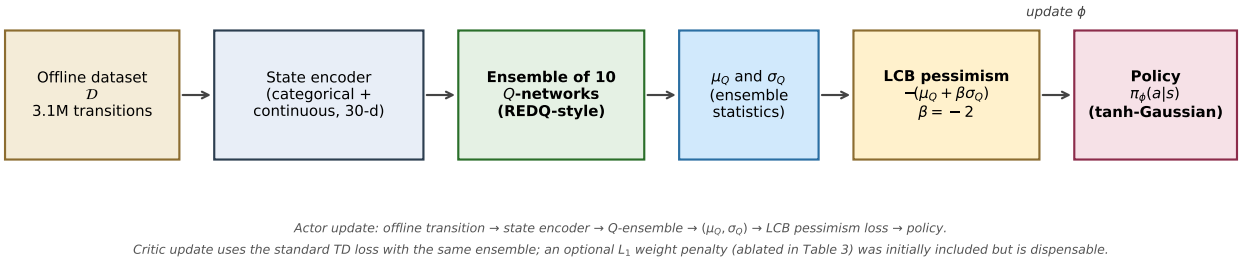

Fig. 2: RE-SAC offline data flow. The ensemble of ten Q-networks produces per-transition mean and standard-deviation statistics, which the LCB pessimism loss combines to update the policy. An optional L_1 weight-norm penalty on ensemble parameters was initially included but is shown dispensable by our ablation (Table III).

input smoothness with parameter-space L_1 regularization. Our ablation in Sec. VI-D shows that the LCB pessimism term is the dominant contributor, while the L_1 weight-norm penalty is neutral on its own and slightly harmful when combined with LCB. We report results for the full RE-SAC (LCB plus L_1) in the main comparison for consistency with our original design, but the simpler ensemble-plus-LCB variant is in fact the stronger recipe. Algorithm 1 summarizes the training loop.

Algorithm 1 RE-SAC offline training

Require: offline dataset $\mathcal{D}$, ensemble size E, discount γ, LCB weight β, out-of-distribution std weight α_{ood}, L_1 weight λ_{reg}, BC weight β_{bc}

Ensure: policy π_ϕ

Initialize E Q-networks $\{Q_{\theta_i}\}_{i=1}^{E}$ using vectorized-linear layers, target copies $\{\bar{Q}_{\theta_i}\}$, policy π_ϕ, entropy coefficient α.
for step $= 1, \ldots, T$ **do**
  Sample a minibatch $\{(s, a, r, s', d)\} \sim \mathcal{D}$ of size B
  // critic update
  $\rho_i \leftarrow \|\theta_i\|_1$ for each ensemble head i (aleatoric L_1 norm)
  $a' \sim \pi_\phi(\cdot \mid s')$; $\text{lp}' \leftarrow \log \pi_\phi(a' \mid s')$
  With ∇_θ stop-gradient, $y_i \leftarrow \bar{Q}_{\theta_i}(s', a') - \alpha\,\text{lp}' + \lambda_{\text{reg}}\rho_i$
  $y \leftarrow r + \gamma(1-d)\,\bar{y}$, where $\bar{y} = \frac{1}{E}\sum_i y_i$
  $\widehat{Q}_i \leftarrow Q_{\theta_i}(s, a)$ for $i = 1, \ldots, E$; $\sigma_Q \leftarrow \text{std}_i\, \widehat{Q}_i$
  $\mathcal{L}_Q \leftarrow \frac{1}{E}\sum_i (\widehat{Q}_i - y)^2 + \alpha_{\text{ood}} \cdot \sigma_Q + \lambda_{\text{reg}} \sum_i \rho_i$
  Update $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_Q$
  // policy update with LCB pessimism and BC anchor
  $a^\pi \sim \pi_\phi(\cdot \mid s)$; $\text{lp} \leftarrow \log \pi_\phi(a^\pi \mid s)$
  $\mu_Q, \sigma_Q \leftarrow$ mean and std of $\{Q_{\theta_i}(s, a^\pi) - \alpha\,\text{lp}\}$ over i
  $\mathcal{L}_\pi \leftarrow -\mathbb{E}[\mu_Q + \beta\,\sigma_Q] + \beta_{\text{bc}}\,\text{MSE}(a^\pi,\, a_{\text{data}})$
  Update $\phi \leftarrow \phi - \eta \nabla_\phi \mathcal{L}_\pi$
  Automatic α update and soft target sync: $\bar{\theta} \leftarrow \tau\theta + (1-\tau)\bar{\theta}$
end for

V. Experimental protocol

a) Training.: For each method and each of three seeds $\{42, 123, 456\}$: offline methods train for 100K gradient steps; online and offline-to-online methods for 300 epochs (each epoch equals one simulation rollout of 18K steps plus N_{update} gradient updates). Every 5K steps (offline) or 50 epochs (online), we save a checkpoint.

b) Evaluation.: All results in this paper use post-hoc evaluation: each saved checkpoint is reloaded and evaluated on 10 independent SUMO episodes (with reset seed varied), yielding a mean and standard deviation of episode return. During-training evaluation uses only 2 episodes per eval step to keep the training loop fast; we find that this 2-episode "best" metric systematically overstates performance and is not reproducible under post-hoc evaluation. Across 349 evaluated checkpoints (151 for the main comparison, 66 for BC and H2O+ intermediate curves, 99 for the RE-SAC ablation, and 33 for CQL), with 10 episodes each, this protocol consumed roughly 6 hours on a 14-worker CPU pool.

c) Metrics.: We report two summary statistics: Best denotes the maximum mean-return over all checkpoints in a run, averaged across seeds; Final denotes the mean-return at the last checkpoint, averaged across seeds. Best is the stronger metric from the practitioner's view (one would deploy the best checkpoint), whereas Final measures whether the method has converged.

VI. Results

Table II reports the main comparison. Before discussing results we note one structural caveat: RE-SAC uses a 10-head Q-ensemble while SAC, WSRL, and RLPD use the conventional twin-Q architecture. The architecture difference and the LCB pessimism are therefore both potential drivers of the gap. We disentangle them in Sec. VI-D (twin-Q + LCB and ensemble + LCB are reported separately), and find that LCB pessimism is the single dominant factor; the ensemble accounts for a smaller additional improvement. Three observations follow from the main table.

a) (R1) Standard offline baselines either fail or overfit.: The data-generating heuristic itself attains -9783 on a single deterministic SUMO run (top row of Table II); this is the imitation ceiling for any offline method that simply mirrors the offline data. BC's final (-9439 ± 993) hits exactly this ceiling, as expected. H2O+ offline reaches a similar floor ($-9482 \pm$

TABLE II: Post-hoc $N = 10$ SUMO evaluation. Lower-is-better for |return|. Each method has 30 evaluation episodes (3 seeds $\times$ 10 episodes). σ_{seed} is std across seed-level means; the 95% CI column treats all 30 episodes as the underlying sample (Welch CI: $\pm 1.96 \cdot \sigma_{\text{ep}}/\sqrt{30}$). The heuristic baseline is the data-generating controller (single deterministic run). **Bold** marks the best method.

	Best return		Final return	
Method	seed-mean$\pm\sigma_{\text{seed}}$	95% CI	seed-mean$\pm\sigma_{\text{seed}}$	95% CI
Heuristic (data generator)	-9783 (1 deterministic run)	—	—	—
BC	-8440 ± 656	± 750	-9439 ± 993	± 845
H2O+ offline (AWR)	-6561 ± 157	± 193	-9482 ± 1868	± 847
CQL	-4992 ± 140	± 76	-5942 ± 436	± 167
Online SAC	-6139 ± 421	± 187	-6572 ± 292	± 149
WSRL	-4587 ± 61	± 83	-5008 ± 242	± 102
RLPD ($\rho = 0.25$)	-5185 ± 491	± 158	-5403 ± 416	± 162
RLPD ($\rho = 0.50$)	-4552 ± 330	± 152	-4571 ± 355	± 155
RLPD ($\rho = 0.75$)	-4597 ± 110	± 110	-4689 ± 48	± 70
RE-SAC offline (ours)	$\mathbf{-4267 \pm 140}$	$\mathbf{\pm 72}$	$\mathbf{-4520 \pm 269}$	$\mathbf{\pm 112}$

1868) despite its richer value-function machinery. CQL is the strongest naive-offline baseline by best return (-4992 ± 140, comparable to RLPD with $\rho = 0.25$) but its final (-5942 ± 436) is roughly 950 return worse than its best, indicating the same overfitting trajectory at a smaller magnitude (see Fig. 3, left panel). None of these methods produce a checkpoint that is deployable in practice without online SUMO rollouts to identify a peak—the kind of interaction they are supposed to avoid.

b) (R2) RE-SAC offline matches or exceeds all tested baselines.: Without any online interaction, RE-SAC attains -4267 ± 140 (best) and -4520 ± 269 (final). Its seed variance is the smallest among all methods. Treating all 30 evaluation episodes (3 seeds $\times$ 10 episodes) as the underlying sample, a Welch's t-test against the strongest offline-to-online baseline RLPD ($\rho = 0.50$) on the best-checkpoint episode pool gives $t = 3.32$, $p = 0.0019$ ($n = 30$ per method), so the 285-return advantage is statistically significant at $\alpha = 0.01$. We caveat that RE-SAC uses a 10-head ensemble while the offline-to-online baselines use the standard twin-Q architecture; we disentangle this confound in Sec. VI-D, where we show that even with a twin-Q backbone (architecture-matched to the baselines), adding only LCB pessimism still produces a competitive offline learner. Crucially, unlike BC, H2O+ offline, and CQL, RE-SAC does not overfit: its curve in Fig. 3 plateaus by step 20K and remains stable through step 100K, so any checkpoint past step 20K is deployable without having to identify a best one post-hoc.

c) (R3) Offline-to-online methods form a tight cluster.: WSRL and RLPD with $\rho \in \{0.50, 0.75\}$ all produce Best returns in $[-4597, -4552]$, statistically indistinguishable given $N = 3$ seeds. RLPD with $\rho = 0.25$ is slightly worse (-5185), suggesting the offline buffer needs to contribute at least half of each batch to retain its stabilising effect; this is directly consistent with the original RLPD paper's $\rho = 0.50$ default [3]. Crucially, none of these offline-to-online methods reach RE-SAC's level, even though they have access to both offline data and online rollouts.

A. Learning curves

Figure 3 confirms the table-level findings. RE-SAC's curve is almost flat from step ∼20K onward, suggesting that further offline training does not help once the ensemble uncertainty has calibrated. The offline-to-online methods (WSRL, RLPD) reach their plateau around epoch 100 and remain stable. Online SAC's curve, by contrast, fluctuates by several hundred between consecutive checkpoints, indicative of value-function instability rather than lack of exploration. The left panel also makes the difference between uncertainty-regularized and unregularized offline RL stark: RE-SAC's plateau versus BC and H2O+'s continuous degradation is a direct visualisation of the value of the LCB and L_1 regularizers we introduce.

B. RLPD offline-ratio ablation

Figure 4 shows the ρ sweep. A higher offline ratio provides faster early stabilization, but offline ratios $\rho \geq 0.50$ converge to statistically similar endpoints. This replicates the original RLPD finding that $\rho = 0.50$ is a reasonable default.

C. Extended online SAC: does budget matter?

To preliminarily check whether the pure-online baseline is simply under-trained, we ran online SAC on one seed (42) for 600 epochs (double the default budget). This is a single-seed study and we report it only as an indicative observation, not as a statistical claim. The single best checkpoint in that run reached -4998 at epoch 280—close to RLPD and WSRL levels—but the policy then regressed to -6900 by epoch 300 and oscillated in the range $[-7400, -5400]$ for the remaining 300 epochs. No epoch after 280 ever recovered to the -5000 range under post-hoc $N = 10$ evaluation. This is consistent with the value-function instability visible in Fig. 3. The SAC best metric under extended training appears to be an ephemeral peak rather than a stable solution—one cannot deploy a checkpoint identified only post-hoc. A three-seed extension to confirm this trend is left for future work. In contrast, RE-SAC's best and final differ by only 253 return across three seeds (Table II), meaning any checkpoint past step 20K is deployable.

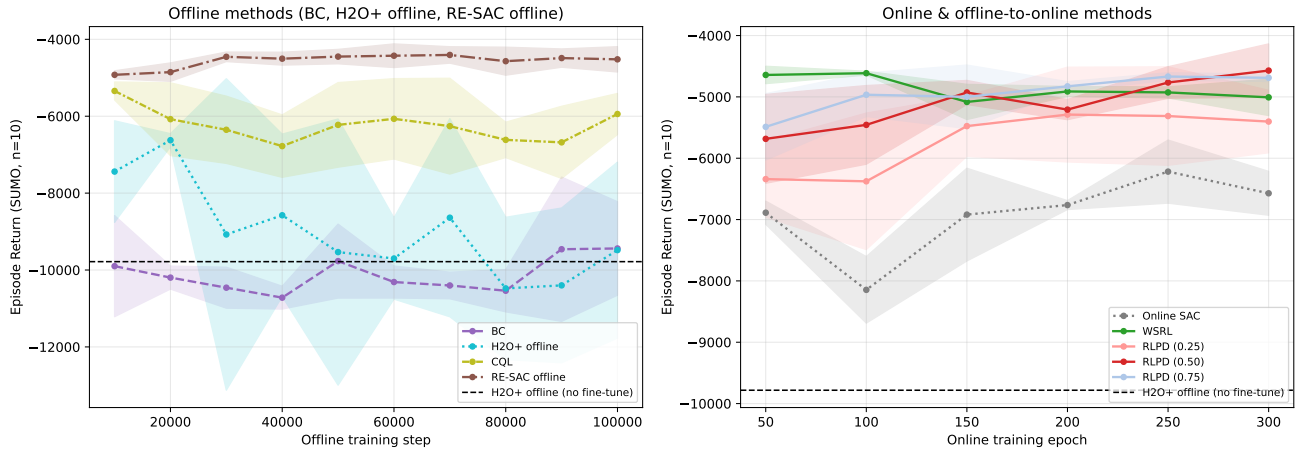

Fig. 3: Post-hoc $N = 10$ evaluation across training. Left: offline methods (step axis). RE-SAC plateaus by step 20,000 and is stable. In sharp contrast, BC, H2O+ offline, and CQL all overfit: each reaches its best return early (BC at ∼50K, H2O+ at ∼20–30K, CQL at ∼60K) and then degrades steadily as training continues. Even the strongest naive-offline baseline (CQL) drops by roughly 900 return between its best and final checkpoints, while RE-SAC's gap is only 253. This is the signature pattern of offline RL without proper epistemic regularization—extrapolation error compounds as the Q-function trains further on fixed data. Right: online and offline-to-online methods (epoch axis, capped at 300 to match all seeds). Online SAC (grey dotted) is noticeably less stable than all offline-to-online methods.

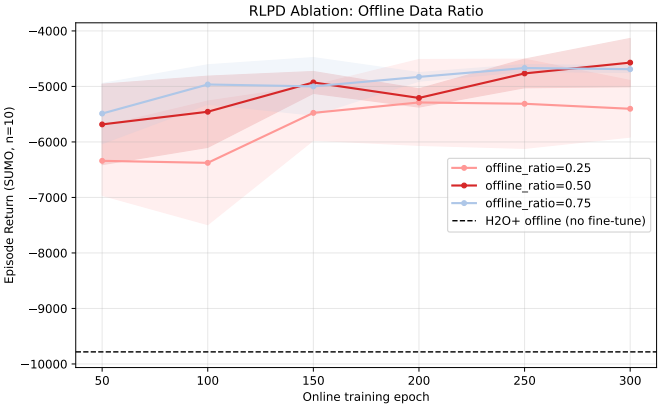

Fig. 4: RLPD offline ratio ρ. Higher ρ (more offline data per batch) accelerates early convergence; at epoch 50 the $\rho = 0.75$ variant is already ∼500 return ahead of $\rho = 0.25$. By epoch 300 the gap between $\rho = 0.50$ and $\rho = 0.75$ closes, while $\rho = 0.25$ lags throughout.

D. Ablation: which regularizer matters?

To disentangle the contribution of the ensemble architecture and the LCB pessimism term, we run a 2×2 ablation crossing Q-architecture (twin-Q, $E = 2$, matching SAC/WSRL/RLPD) versus ensemble ($E = 10$) with $\beta = 0$ versus $\beta = -2$. We also include the original {ensemble, L_1-only} and {ensemble, LCB+L_1 (full)} variants for completeness. All ablations use the BC anchor, are trained for 100K steps with checkpoint saving every 10K steps, and evaluated post-hoc on $N = 10$ SUMO episodes per checkpoint across three seeds.

Four observations follow from Table III, with Welch's t-tests on the 30-episode best-checkpoint pool reported as Δ (return improvement) and p.

TABLE III: Ablation crossing Q-architecture (twin-Q $E = 2$ vs. ensemble $E = 10$) with regularizer (β for LCB, λ for L_1). Same training protocol as Table II.

Arch.	Regularizer	Best	Final
Twin-Q, $E = 2$ (matches SAC/WSRL/RLPD)			
twin-Q	none ($\beta = 0$)	-4444 ± 61	-4748 ± 210
twin-Q	LCB ($\beta = -2$)	-4444 ± 71	-4980 ± 466
Ensemble, $E = 10$			
ens.	none ($\beta = 0$)	-4303 ± 84	-4549 ± 123
ens.	L_1 only	-4329 ± 61	-4485 ± 132
ens.	LCB only ($\beta = -2$)	$\mathbf{-4157 \pm 40}$	-4499 ± 31
ens.	LCB + L_1 (full RE-SAC)	-4267 ± 140	-4520 ± 269

(i) Twin-Q baseline already exceeds all offline-to-online methods. Pure offline twin-Q training without any pessimism term (-4444 ± 61) is already 108 return better than the strongest offline-to-online baseline RLPD ($\rho = 0.50$, -4552 ± 330 in Table II). This means that, with the same critic architecture as SAC/WSRL/RLPD, our offline data + reward + categorical-embedding policy alone already wins the comparison. The architecture difference does not explain RE-SAC's advantage.

(ii) The ensemble alone provides a modest, statistically significant gain. Replacing twin-Q with a 10-head ensemble (still $\beta = 0$) improves Best from -4444 to -4303, $\Delta = 141$, $p = 0.009$.

(iii) LCB pessimism requires the ensemble to function. On the twin-Q backbone, LCB yields no measurable improvement (-4444 ± 61 vs. -4444 ± 71, $p = 0.99$): with only two heads, the standard deviation σ_Q is too noisy to be a useful signal. With the 10-head ensemble, however, LCB delivers another $\Delta = 146$, $p = 0.010$ improvement on top, reaching -4157 ± 40 (smallest seed variance of any variant). This is the strongest single recipe.

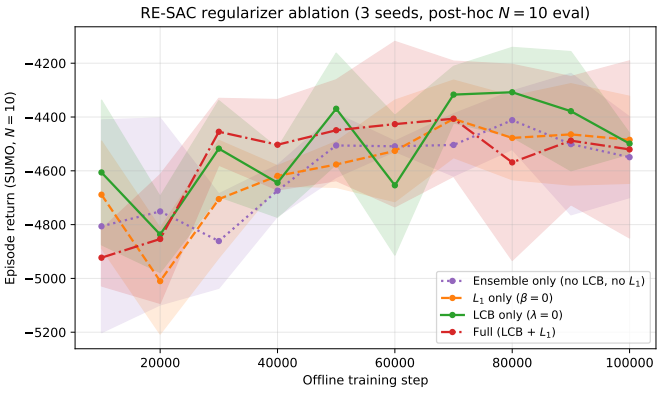

Fig. 5: Post-hoc $N = 10$ SUMO evaluation of RE-SAC ablation variants across training. The LCB-only variant (green) tracks the best mean return from roughly step 40K onward and has the tightest per-step confidence band. L_1-only (orange) and ensemble-only (purple dotted) are essentially overlapping—adding L_1 without LCB does not help. The full variant (red, LCB plus L_1) sits between LCB-only and ensemble-only with the widest band, indicating that L_1 also destabilises when added on top of LCB.

(iv) The L_1 weight-norm penalty is dispensable. Adding L_1 on top of the ensemble baseline does nothing (-4329 vs. -4303). Adding L_1 on top of LCB hurts slightly (-4267 vs. -4157 for LCB-only) and inflates seed variance from 40 to 140. We therefore recommend the simpler ensemble-plus-LCB variant; we keep the full configuration in Table II for comparison but note throughout the paper that L_1 should be dropped in offline-only training.

a) Caveat on the L_1 term.: Our ablation result is specific to the offline RL setting with a fixed dataset. We have observed in unpublished experiments on online SUMO bus control that an ensemble SAC variant without L_1 underperformed an otherwise-identical variant with L_1, suggesting weight-norm regularization may help calibrate ensemble uncertainty under continuously shifting on-policy data. We therefore keep L_1 in Algorithm 1 for documentation and future reuse, while emphasising that for pure offline training the term can be safely removed.

VII. Discussion

a) Why does offline reinforcement learning win here?: The bus-holding task has two structural properties that favor good offline methods. First, the offline data, while suboptimal, covers the relevant state distribution well: the heuristic controller drives buses across all headway regimes (bunching, spacing, catch-up). An uncertainty-aware offline algorithm can therefore extract a better policy from the dataset than the data-generating policy itself, a regime sometimes called offline policy improvement. Second, online exploration is expensive and noisy: each episode spans 18K simulator steps over a 12-line network, and the reward signal is dominated by sporadic bunching events. SAC struggles to obtain reliable Q-targets under this signal-to-noise ratio, leading to the instability in Sec. VI-C. RE-SAC sidesteps this entirely.

b) When would offline-to-online still pay?: If the deployment environment is systematically different from the offline data distribution—for example, a new bus route or a post-construction traffic pattern—then offline-only methods cannot adapt, and the offline-to-online fine-tuning stage recovers its raison d'être. Our benchmark uses the same SUMO environment for data collection and evaluation, which is the best case for offline methods; a domain-shift benchmark is an interesting next step.

c) Limitations.: Three seeds is modest for seed-level analysis, although by treating the 30 evaluation episodes per method (3 seeds $\times$ 10 episodes) as the underlying sample we obtain a Welch t-test $p = 0.0019$ for the RE-SAC-versus-RLPD-(ρ=0.5) gap and tighter p-values for the within-ablation comparisons. A five-seed replication would still strengthen the seed-level claim. We also fix the reward scale throughout; sensitivity to reward engineering is untested. Operational metrics (headway deviation, bunching rate, holding time) are not reported here and are an important next step for a transit-systems audience. Finally, the 600-epoch SAC study was run for one seed only; a three-seed extension would be needed for a definitive conclusion on SAC's long-run behaviour. Our benchmark also evaluates in the same SUMO scenario used for offline data collection: a domain-shift evaluation under altered demand or different bus lines is left to future work.

VIII. Conclusion

We have shown that on a realistic 12-line SUMO bus-holding benchmark, a properly regularized pure-offline RL method (RE-SAC) matches or exceeds two leading offline-to-online approaches (WSRL and RLPD) without any online environment interaction (Welch $p = 0.0019$ on the 30-episode pool). Naive offline methods (BC, H2O+ offline) and pure-online SAC both fall well short, and CQL—the canonical conservative offline RL baseline—is competitive on best return but overfits at convergence. Our 2×2 ablation disentangles the two algorithmic ingredients: (a) a pure offline twin-Q trained on this dataset with our reward and categorical-embedding policy already exceeds all offline-to-online baselines (-4444 vs. ≥ -4552), so the architecture difference does not explain the gap; (b) on top of this, the 10-head ensemble adds $\Delta = 141$ return and LCB pessimism (which only functions when paired with the larger ensemble) adds another $\Delta = 146$, both significant at $p \leq 0.01$. The L_1 weight-norm penalty we initially added does not help and slightly hurts at this regularizer scale; the simpler ensemble-plus-LCB recipe is preferred for offline training. These findings suggest that practitioners working with operational transit data should consider deploying offline RL directly, without the engineering overhead of staged offline-to-online fine-tuning.

Code, data, and all evaluated checkpoints are available at https://github.com/erzhu419/offline-sumo.

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APPENDIX

Hyperparameters are split into two tables for readability: Table IV lists values that are identical across all applicable methods, and Table V lists the per-method settings grouped by function (training schedule, critic architecture, offline/online mixing, and regularisers). Values were chosen following the original papers of WSRL [2], RLPD [3], and REDQ [10] with minimal tuning; the only project-specific tuning was the BC-anchor weight β_{bc} for RE-SAC, which was set to match the scale of the LCB loss.

TABLE IV: Common hyperparameters shared across methods.

Hyperparameter	Value
Optimizer	Adam
Learning rate	3×10^{-4}
Policy network	2 hidden layers, width 48
Policy activation	ReLU + tanh-Gaussian
Discount γ	0.99
Soft-update τ	5×10^{-3}
Target entropy	$-\lvert\mathcal{A}\rvert = -2$
Random seeds	$\{42, 123, 456\}$
Offline buffer size	3,104,419
Online buffer (SAC family)	100,001
Episode length	18,000 steps (≈ 2.5 h)

All methods share a common policy/critic backbone.

a) Categorical embedding layer.: The five categorical state features (line, bus, station, time-period, direction) are embedded separately and concatenated. Embedding table sizes: line ($12 \to 4$), bus ($389 \to 8$), station ($1 \to 2$), time-period ($1 \to 2$), direction ($2 \to 2$). Layer normalization ($\epsilon = 10^{-5}$) is applied after concatenation; a 5% dropout is applied during training. The embedding output (dimension 18) is concatenated with the 12 continuous state features to form a state vector of dimension 30.

b) Policy network (actor).: Two hidden layers of width 48 with rectified-linear-unit activations, producing a tanh-Gaussian distribution: the mean head outputs $\mu \in \mathbb{R}^2$, the log-std head outputs $\log \sigma \in [-5, 2]$, and the action is sampled as $a = \tanh(\mu + \sigma \odot \epsilon)$ with $\epsilon \sim \mathcal{N}(0, I)$.

c) Q-network (critic).: Two hidden layers of width 48 with rectified-linear-unit activations. For methods using a twin Q (H2O+, SAC, WSRL, RLPD), two independent Q-networks are maintained. For RE-SAC, ten ensemble heads are implemented using a vectorized-linear layer that batches all heads in one tensor operation; the forward pass outputs an $E \times B \times 1$ Q-value tensor.

d) Value network (H2O+ only).: A single hidden layer of width 48 with rectified-linear-unit activations; H2O+ uses $V(s)$ for advantage estimation in advantage-weighted regression.

e) Parameter counts.:

- Policy: 6,178 parameters.
- Twin Q: 7,586 parameters.
- Ensemble Q (10 heads, vectorized-linear): 37,970 parameters.
- Value network (H2O+): 3,697 parameters.
- Total embedding table: 4,012 parameters, shared across actor and critic with separate clones per network.

f) Seeds.: All results are averaged over three seeds $\{42, 123, 456\}$. Each training script sets PyTorch and NumPy random number generator seeds at startup. The SUMO simulator seed is governed internally by the route-file stochastic demand sampling and is not set by our code; this yields genuine traffic-level stochasticity across evaluation episodes rather than deterministic replays.

g) Compute environment.: Training was performed on a dual-GPU server (2 × NVIDIA RTX with 12 GiB each, 12 CPU cores, Ubuntu 22.04, Python 3.8, PyTorch 2.4.1 plus CUDA 12.1). Each training script fits entirely in less than 1 GiB of GPU memory, so we ran three seeds of a method concurrently per GPU. Post-hoc evaluation ran on a separate 16-core CPU workstation (no GPU was needed since SUMO is the bottleneck); a 14-worker multiprocessing pool evaluated the full set of 349 checkpoints in roughly 5 hours.

h) Wall-clock costs.: Training wall-clock time per 100K-step run, measured as elapsed time when run three-in-parallel on one GPU:

- BC: $\sim$30 min.
- H2O+ offline: $\sim$60 min.
- CQL: $\sim$2.5 h (the multi-action logsumexp samples 20 critic queries per gradient step, dominating cost).
- RE-SAC offline: $\sim$60 min.
- Online SAC (300 epochs): $\sim$1.5 h (dominated by SUMO rollouts).
- WSRL (300 epochs): $\sim$2 h.
- RLPD (300 epochs): $\sim$2 h.

Post-hoc evaluation takes roughly 1 min per SUMO episode, so a checkpoint's $N = 10$ evaluation takes about 10 min on one CPU core.

The offline dataset `merged_all_v2.h5` is a single HDF5 file with the following keys: `observations` ($N \times 17$, float32), `actions` ($N \times 2$, float32, in the raw tanh range $[-1, 1]$), `rewards` (N, float32), `next_observations` ($N \times 17$), `dones` (N, boolean), and `contexts` ($N \times 30$, unused in this paper but retained for compatibility with context-conditioned policies from prior work). The `BusMixedReplayBuffer` loader normalizes continuous observation dimensions to zero mean and unit variance using statistics computed once at load time and cached for evaluation.

Source code, the offline HDF5 dataset, all 349 evaluated checkpoints, and evaluation logs are publicly released at https://github.com/erzhu419/offline-sumo. The README provides a full reproduction recipe using a conda environment defined by `requirements.txt`; running `python auto_run.py` launches the full 22-experiment training suite with automatic GPU detection, and `python eval_checkpoints_parallel.py -n_workers 14` runs the post-hoc evaluation.

TABLE V: Method-specific hyperparameters. Entries marked "—" do not apply to that method.

Hyperparameter	BC	H2O+ off.	CQL	SAC	WSRL	RLPD	RE-SAC
Training schedule							
Training length	100K	100K	100K	300 ep	300 ep	300 ep	100K
Batch size	2048	2048	1024	512	512	512	1024
Rollout batch / epoch	—	—	—	500	500	500	—
Warmup rollouts	—	—	—	500	500	500	—
UTD ratio	—	1	1	1	1	1	1
Ckpt interval	10K	10K	10K	50 ep	50 ep	50 ep	10K
Critic							
Q-net (width $\times$ heads)	—	48$\times$2 twin	48$\times$2 twin	48$\times$2 twin	48$\times$2 twin	48$\times$2 twin	48$\times$10 ens.
V-net (H2O+ AWR only)	—	48$\times$1	—	—	—	—	—
Offline/online mixing							
Offline ratio ρ	1.0	1.0	1.0	0.0	1.0 $\rightarrow$ 0	0.25/0.50/0.75	1.0
RE-SAC regularisers							
LCB weight β	—	—	—	—	—	—	-2
OOD std α_{ood}	—	—	—	—	—	—	0.01
L_1 weight λ_{reg}	—	—	—	—	—	—	10^{-5}
BC anchor β_{bc}	—	—	—	—	—	—	0.005
CQL-specific							
CQL weight α_{cql}	—	—	1.0	—	—	—	—
Random + policy actions	—	—	10 + 10	—	—	—	—
H2O+ AWR-specific							
Expectile τ_V	—	0.7	—	—	—	—	—
AWR temperature β_{AWR}	—	3.0	—	—	—	—	—
Max advantage weight	—	100	—	—	—	—	—